

Supplementary Material:

Evaluating the Impact of Video Compression on Deep Learning Performance in Surgical Video Analysis

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S1. OVERVIEW

This document provides supplementary material for the main paper, including dataset visualizations, qualitative compression examples, and accuracy-based performance matrices that complement the macro F1 results in the main paper. All experimental settings, architectures, and training procedures follow those described in the main paper.

S2. DATASET VISUALIZATIONS

S2.1. Phase Imbalance

Figure S1 reports the mean fraction of surgery time occupied by each phase, averaged across all 80 videos. Calot Triangle Dissection and Gallbladder Dissection together account for roughly 75% of surgical duration, while Clipping/Cutting and Gallbladder Retraction each occupy under 5%. This severe imbalance motivates the use of macro F1 as the primary evaluation metric in the main paper, since accuracy is dominated by performance on the majority phases.

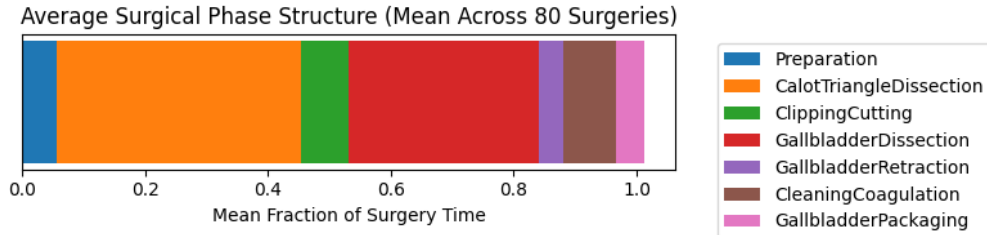


Figure S1: Mean fraction of surgery time per phase, averaged across the 80 Cholec80 videos. Two phases (Calot Triangle Dissection, Gallbladder Dissection) dominate the dataset, while several clinically important phases (Clipping/Cutting, Gallbladder Retraction) are brief.

S2.2. Tool-Phase Co-occurrence

Figure S2 shows the conditional probability of each surgical tool being present during each phase. The strong structure—e.g., Hook dominates Calot Triangle Dissection and Gallbladder Dissection, while Clipper appears almost exclusively during Clipping/Cutting—reflects the procedural workflow and provides the inductive bias exploited by multi-task models such as MTRCNet-CL, which jointly learns tool presence and phase recognition via a correlation loss.

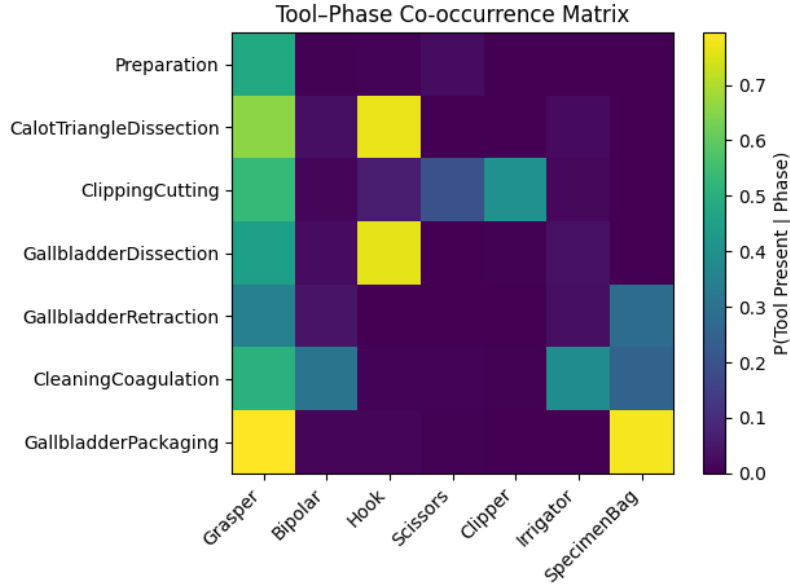


Figure S2: Tool–phase co-occurrence matrix, computed as $P(\text{tool present} \mid \text{phase})$ across Cholec80. Strong tool–phase correspondences explain why multi-task frameworks (e.g., MTRCNet-CL) leverage joint supervision.

S2.3. Surgery Duration

Figure S3 shows the distribution of total surgery duration across the dataset. Most procedures fall between 20 and 50 minutes, with a long tail extending beyond 90 minutes. This variability in procedure length affects the temporal context available to clip-based and TCN-based models.

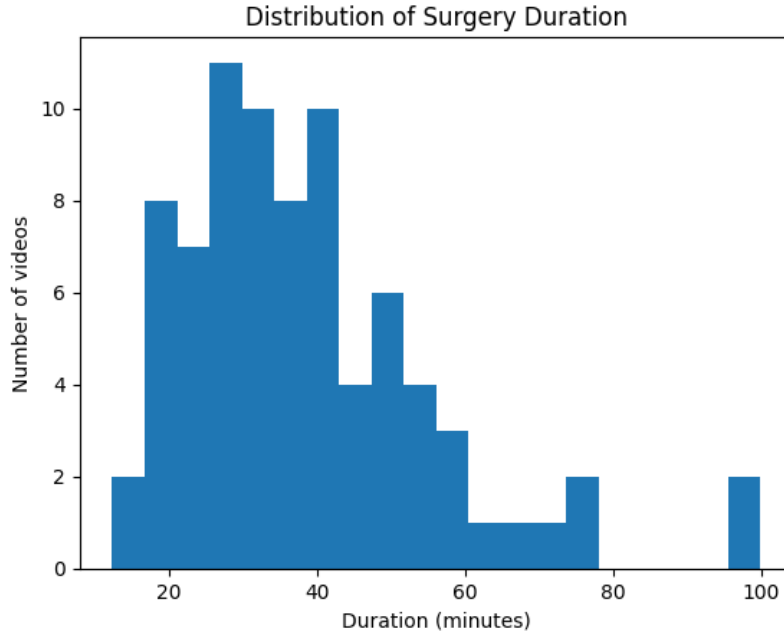


Figure S3: Distribution of total surgery duration across the 80 Cholec80 videos. Most procedures fall between 20–50 minutes, with a heavy right tail.

S2.4. Compression Visual Examples

Figure S4 shows the same surgical frame re-encoded at six CRF levels. Through CRF 35, frames remain visually faithful with only mild loss of fine texture. At CRF 51, prominent block artifacts emerge—tissue boundaries become coarse and the surgical

tool’s metallic highlights are quantized into uniform patches. This pattern is consistent with the Laplacian variance collapse and edge IoU drop reported in the main paper.

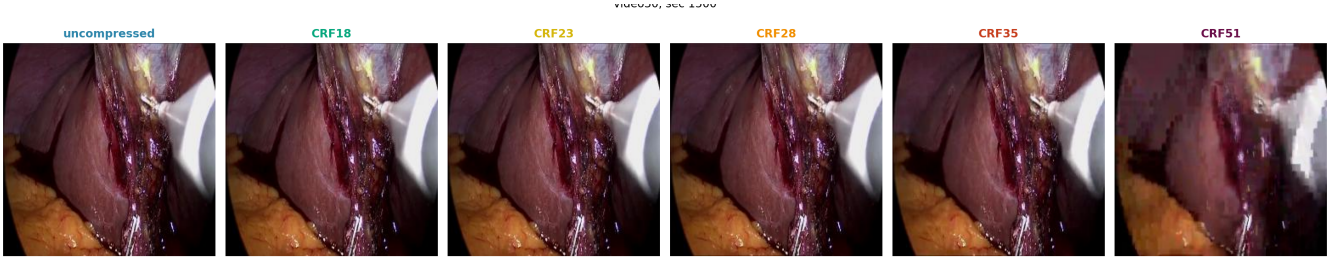


Figure S4: Single surgical frame (video 30, second 1500) re-encoded at increasing CRF levels. The main paper evaluates CRF 23, 28, 35, and 51; CRF 18 and the uncompressed reference are included here for comparison. Visual degradation is minimal through CRF 35 and becomes pronounced at CRF 51.

S3. ACCURACY MATRICES

For completeness, Figure S5 reports train-vs-test compression accuracy across all six architectures. As discussed in the main paper, accuracy is inflated by performance on dominant phases in Cholec80, which is why macro F1 was selected as the primary metric. The accuracy matrices follow the same train-vs-test trends as the F1 results: stable performance through CRF 35, clear degradation at CRF 51, and partial recovery when models are trained directly on CRF 51.

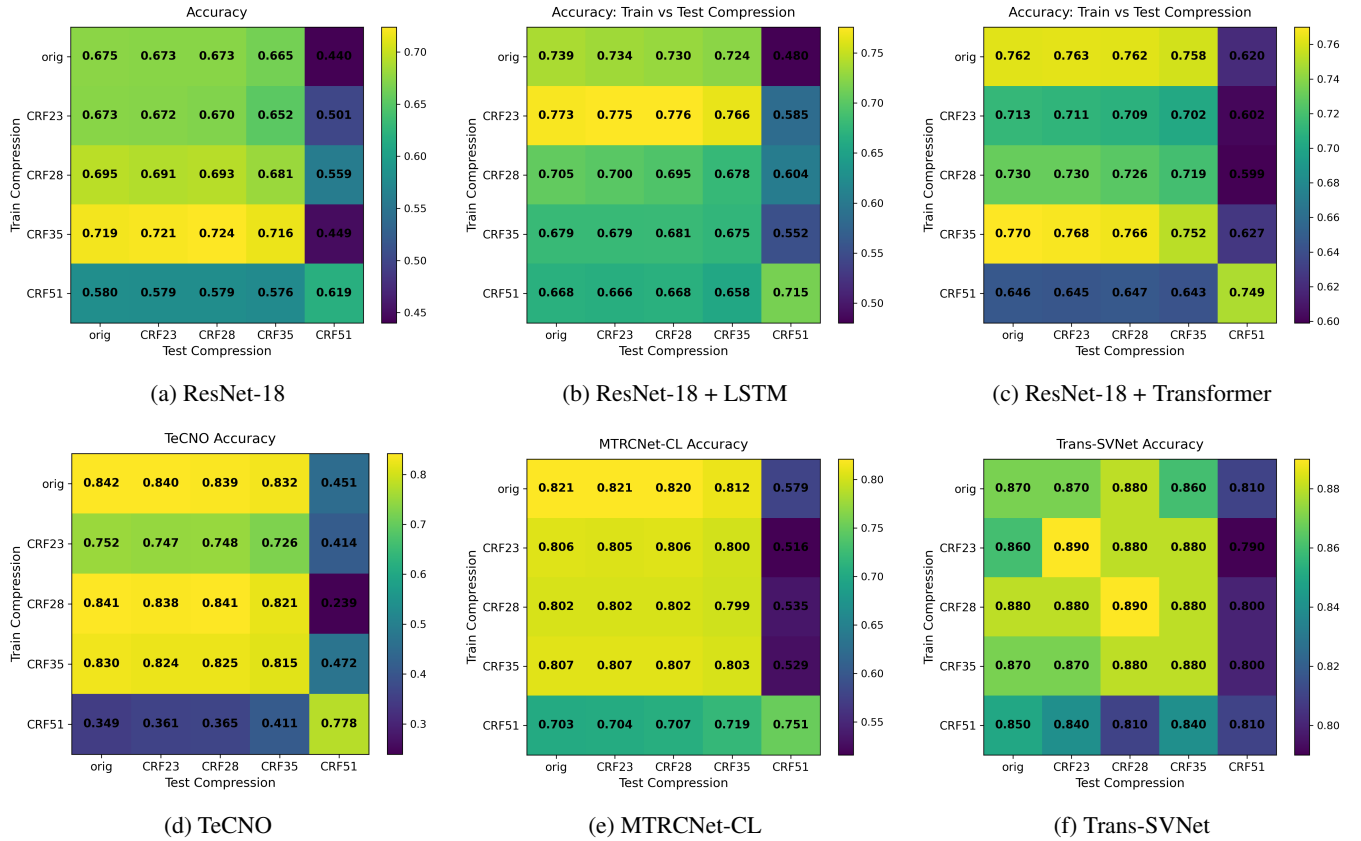


Figure S5: Train-vs-test compression accuracy matrices across all six architectures (top: pilot models; bottom: state-of-the-art). Rows = training CRF; columns = test CRF. Patterns mirror the macro F1 results in the main paper.